Pathways to Power: A Network Analysis of Educational Pipelines in the KSE-500

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Abstract

This study investigates the structural relationship between corporate leadership and educational institutions in Pakistan. We analyzed a bipartite network of 298 KSE-500 CEOs and their 171 alma maters, while prioritizing the Institution-Institution network derived from it to map the "feeders" and "bridges" of corporate power. The analysis reveals a sparse, scale-free network dominated by local super-hubs like the Institute of Business Administration (IBA) and Punjab University. While the network exhibits "Small World" properties, it is fragmented into distinct regional power blocs (Karachi vs. Lahore), with international institutions acting as critical bridges in an otherwise insular educational landscape.

Keywords: Social Network Analysis, Corporate Elites, Centrality, Community Detection, KSE-500.

1 Introduction

Corporate leadership in Pakistan is rarely a product of meritocracy alone; rather, it is deeply embedded in historic social structures. While the dominance of family dynasties is well-documented [2], the mechanisms that sustain this exclusivity are evolving. While the role of biological kinship in propagating this exclusivity has been studied extensively [4], the role of educational institutions as distinct paths or 'pipelines' to power remains underexplored in the Pakistani context. This paper addresses that gap by quantitatively mapping the CEO-Education network of the KSE-500, and investigates universities as alternative and significant pathways to the corporate suite.

Our primary objective is to analyze the **Institution-Institution** network to identify which universities act as hubs where corporate talent from diverse sources can interact, and which universities serve the vital function of connecting disparate corporate circles. By applying Social Network Analysis (SNA) to this network, we also uncover the specific institutional pathways that define Pakistan's corporate architecture along regional and methodological lines. We briefly examine the CEO network to gauge how these institutional structures manifest into social cohesion, though our focus remains on the institutional landscape that shapes Pakistan's corporate elites.

2 Literature Review

This paper builds upon the foundations of Hambrick's [1] "Upper Echelons Theory", which argues that executives' educational backgrounds heavily influence strategic decision-making. Nawaz [5] adapts this lens to Muslim-majority markets, highlighting how the distinct mindset cultivated by degrees like MBAs shapes financial decision-making. Simultaneously, Verhoeven et al. [7] demonstrate how elite institutions act as engines of social reproduction, using "school-linking processes" to control access to power. They argue that rather than nurturing talent equitably, elite education offers a "justification" for privilege, effectively guarding access to power. This is extremely relevant to Pakistan, where Haque and Husain [2] characterize the financial sector as a "small club" dominated by 31 families, prompting the question of whether universities challenge or reinforce these dynastic monopolies.

Methodologically, we employ Social Network Analysis (SNA), drawing on Wasserman and Iacobucci's [8] foundational work on bipartite graphs. Critically influential is Semenova [6]'s multi-method SNA of German executive networks. Beyond just identifying "Small World" properties, Semenova highlights the structural importance of "brokers", i.e. specific actors or subgroups (Like automotive CEOs) that bridge otherwise disconnected cliques. Similarly, Josefy et al. [3] define "elite pipelines" not merely as hiring preferences, but as structural mechanisms driven by "transitivity", where networks tighten through mutual friends, and "homophily", where elites hire those who mirror their own image, creating closed loops that are difficult for outsiders to penetrate.

In the South Asian context, Naudet and Dubost [4] provide a critical counter-narrative to globalization. They argue that Indian corporate networks have actually "densified" through caste and regional ties post-liberalization, suggesting a conscious resistance to meritocratic organization. While such interlocks are documented in India and the West, there is a lack of quantitative SNA scholarship on the Pakistani corporate structure. Our paper fills that gap by evaluating if the KSE-500 mirrors global "Small World" trends or operates on unique, localized dynamics defined by this regional resistance to liberalization.

3 Data Collection and Preparation

Since a dataset of CEOs and their educational institutions was not readily available, we had to curate it by web-scraping the Pakistan Stock Exchange (PSX) and StockAnalysis.com for KSE-500 companies. CEO educational backgrounds were manually extracted from LinkedIn, Market Screener, and annual reports.

We encountered a major challenge in the form of incomplete information, as a significant number of CEOs did not have their educational information publicly available, while others had incomplete information, for example, having their Graduate institution listed but not their Undergraduate institution. After curating information about 397 CEOs on the KSE-500, we deemed that 298 of them were suitable to be included in our dataset as they had sufficiently met our criteria for verifiable and complete educational history. Furthermore, we had to address significant name ambiguity by normalizing institution names.

Ultimately, we constructed a **Bipartite Graph** consisting of **298** CEO nodes and **171** Institution nodes, connected by **438** educational edges. This was projected into two unipartite networks:

- **Institution-Institution**: Connections based on shared alumni. This network consisted of **119** Institution nodes and **124** edges.
- **CEO-CEO**: Connections based on shared alma maters. This network consisted of **238** CEO nodes and **1924** edges.

4 Methodology

We employed the following metrics to analyze network topology:

- **Network Metrics**: Graph Density, Average Path Length, Clustering Coefficient.
- **Centrality**: Degree, Betweenness, and PageRank/Eigenvector.
- **Community Detection**: The Fast Greedy algorithm was used to identify institutional power blocs. Label Propagation was used to identify CEO communities. The community detection algorithms were chosen based on the highest modularity.
- **Theoretical Models**: We compared the empirical network against Erdos-Renyi, Barabasi-Albert, and Watts-Strogatz models to validate structural properties like preferential attachment and "small world".

5 Results

5.1 Bipartite Network

This Bipartite network between CEOs and the Institutions which they have attended is an **undirected graph** consisting of a total of **468** nodes. The nodes include "CEO Nodes" (297) and "Institution Nodes" (171), with the CEO to Institutions ratio being **1.74: 1**.

The bipartite network is defined by extreme sparsity (graph density: 0.004), a feature driven primarily by the low connectivity of actors (CEOs). The average degree for CEO nodes is just 1.475, with 58.2% CEOs holding a single educational affiliation and only 5.7% holding three. In contrast, the average Institution degree is significantly higher at 2.56. This disparity indicates that the network is **institution-driven** rather than actor-driven. Consequently, the connectivity of the projected CEO network depends almost entirely on large, central institutional hubs. In such a sparse environment, shared educational affiliations represent statistically rare and deliberate structurally critical bonds.

Analyzing the bipartite network, we see that 64% of Institutions have only produced a single CEO (Figure 1).

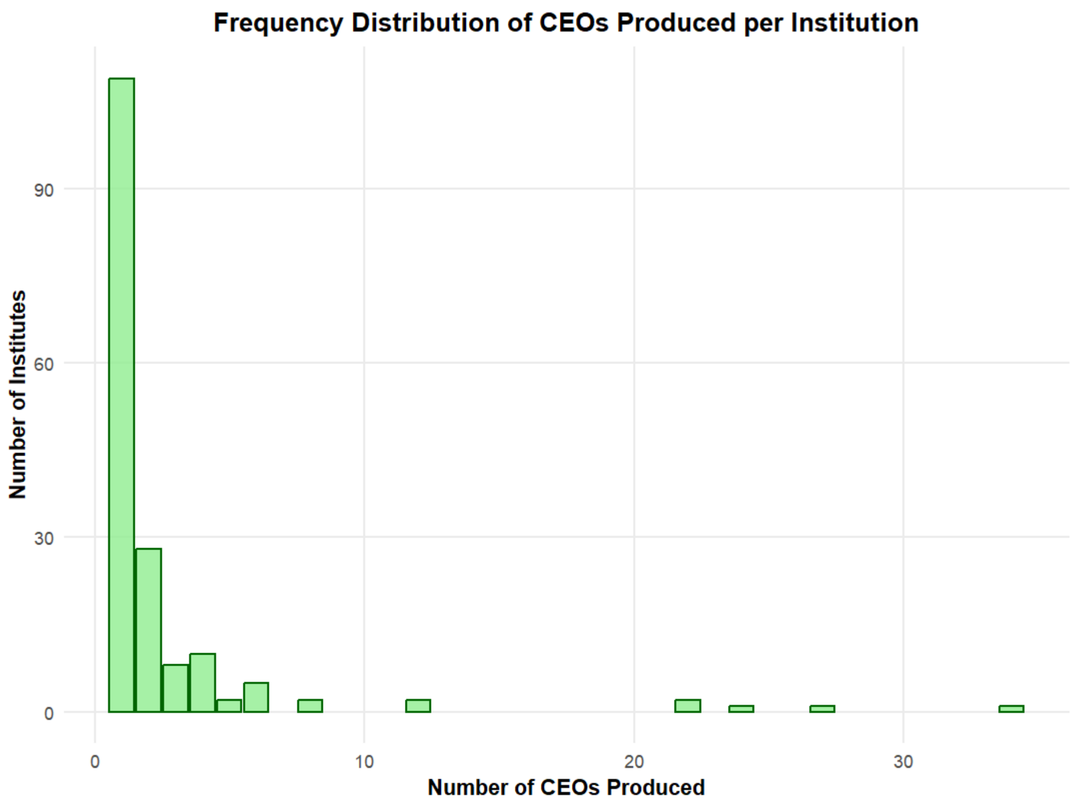

Figure 1: Distribution of CEOs across institutions in the KSE-500 bipartite network, highlighting the dominance of major feeder universities.

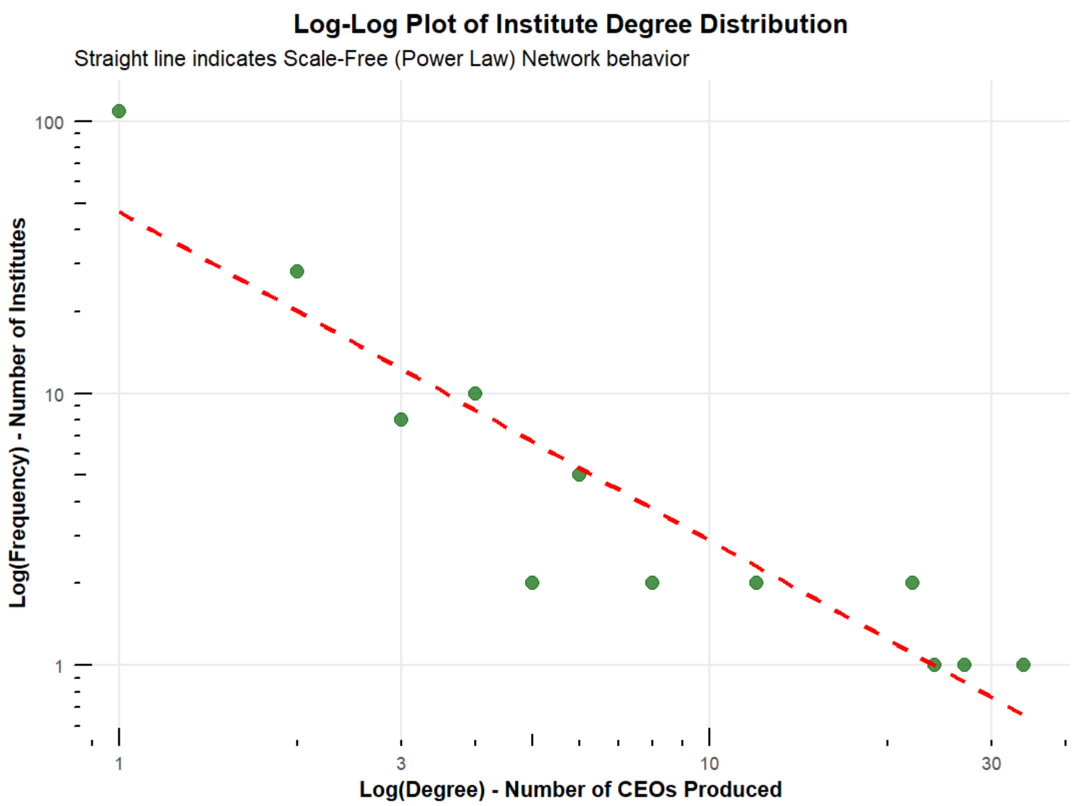

Figure 2: Log-log plot of the Institution Degree Distribution, demonstrating the scale-free nature and preferential attachment in the network.

As evidenced by the extreme right-skewness of the Institution Degree Distribution, the bipartite network seems to show **Scale-free** properties, with a very small number of Institutions (Hubs) having an extremely high degree. This translates to a few educational institutions producing a disproportionately high number of CEOs.

The Scale-free nature of the network is also confirmed by the Log-Log plot of Institute Degree Distribution which follows Power Law (Figure 2). This indicates Preferential Attachment, i.e. **"Rich-get-Richer"** behavior, where new CEO nodes may prefer to attach to "feeder" Institutional nodes with high degree, i.e. educational institutions that have already produced a robust network of CEO alumni. These properties will be further explored in Comparison with **Seminal Network Models**.

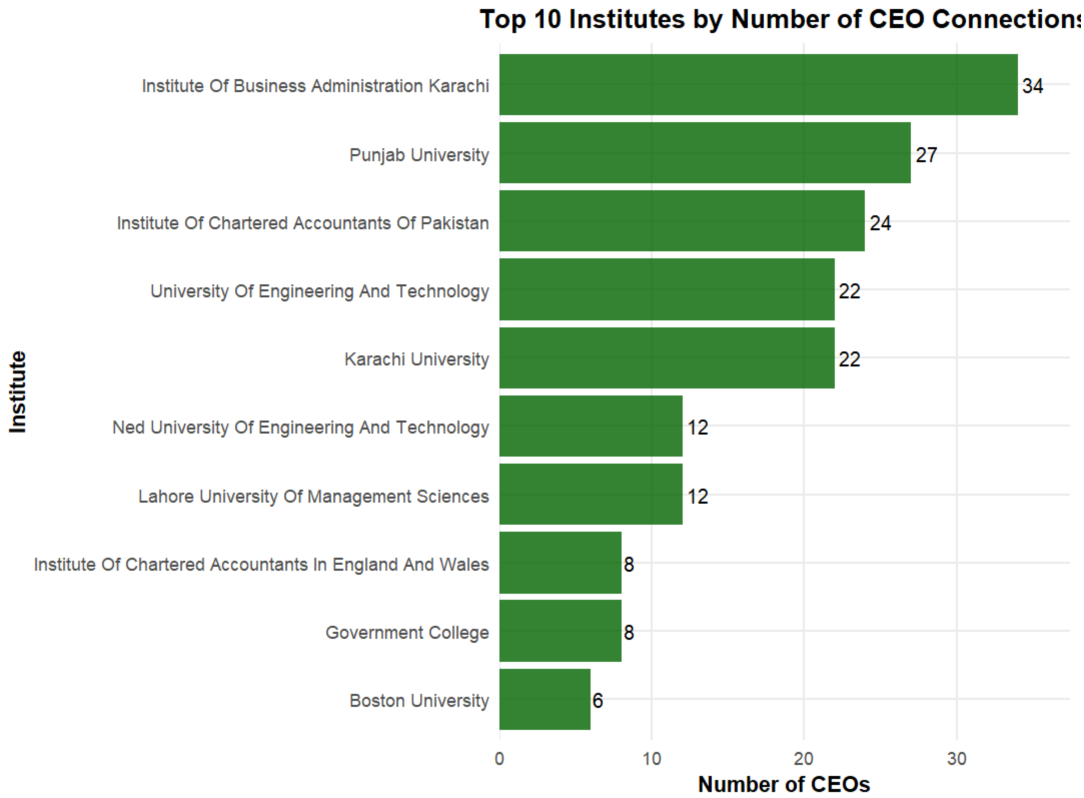

Figure 3: Top 10 institutions by degree in the KSE-500 bipartite network.

The bar graph above represents the top 10 Institute nodes by degree, i.e. the institutions responsible for producing the most CEOs in our network (Figure 3). As evident, the **Institute of Business Administration** (IBA) is the greatest feeder institute in our network. The bar graph also supports our assertion that the network is scale-free as the number of CEOs produced falls sharply from IBA (34) to Boston University (6). Most interestingly, the top 10 institutions are almost exclusively domestic (8 out of 10) and are primarily located in **Karachi** and **Lahore**. This suggests that the core of the KSE-500 network is produced indigenously and is geographically concentrated to Pakistan's two largest cities. This indicates that network sparsity amplifies Preferential Attachment, fueling the aforementioned "Rich-get-Richer" dynamic where visible hubs act as magnets. This implies that aspiring CEOs prioritize established CEO alumni networks over purely academic factors, thus treating institutional choice as a strategic entry point into the corporate elite.

5.2 Institution-to-Institution Projection

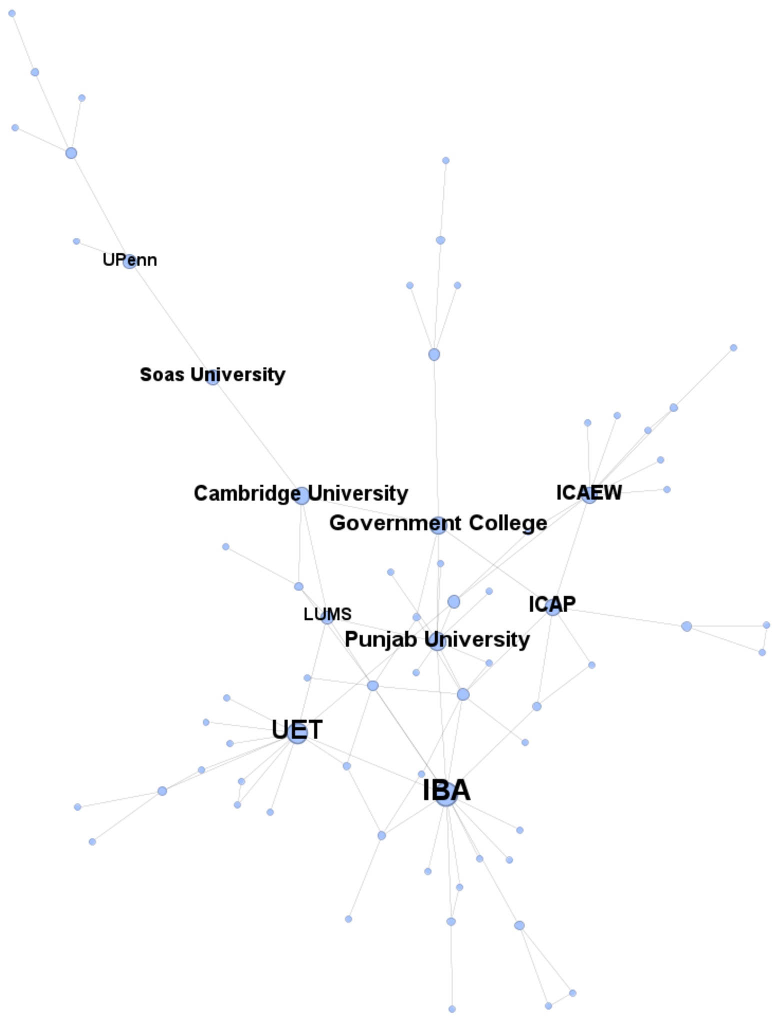

Figure 4: Giant Component of the Institute-Institute Projection.

The Institution-Institution projection represents the academic collaboration network as defined by shared alumni (Figure 4). Two institutions are connected if at least one CEO graduated from them both. The edges are weighted, however, most edges have a weight of 1.

The network is characterized by a low graph density of **0.176**, indicating a sparse structure. The average degree is **2.084**, which aligns with the bipartite finding that most CEOs possess only 1-2 degrees. This low connectivity results in a network diameter of **11** and an average path length of **4.6**. This suggests that while institutions are locally connected, there are significant structural distances between disparate clusters, making the network vulnerable to fragmentation if key bridges are removed.

The most defining characteristic of this network is the disparity between its clustering coefficients. The Global Clustering Coefficient is low **(0.112)**, while the Average

Local Clustering Coefficient is significantly higher **(0.38)**. This indicates that the corporate education landscape for the KSE-500 is fragmented and composed of distinct "islands" or cliques with small-world properties. Information may flow easily within these tight-knit clusters but could face significant bottlenecks when attempting to cross between them.

5.2.1 Community Analysis

Figure 5: Communities within the Institute-Institute Projection.

For further analysis, we isolated the giant component of the network which consists of approximately **60%** of the total nodes. Upon running the Fast Greedy community detection algorithm on the giant component, we discovered that the component is divided into **8** communities (Figure 5).

We identified the following four primary power blocs:

Cluster A: Tech-Business Nexus

Key Hubs: IBA, NED University, NUST, FAST

This cluster represents the modern technocratic elite (Figure 6). It forms a tight nexus between Pakistan's premier business school (IBA) and Pakistan's elite engineering institutions (NED, NUST). The cluster indicates a distinct career pathway, primarily dominated by Karachi-based institutions, where CEOs acquire a STEM-based (Commonly Engineering) education and then proceed to acquire management training, primarily in the form of an MBA.

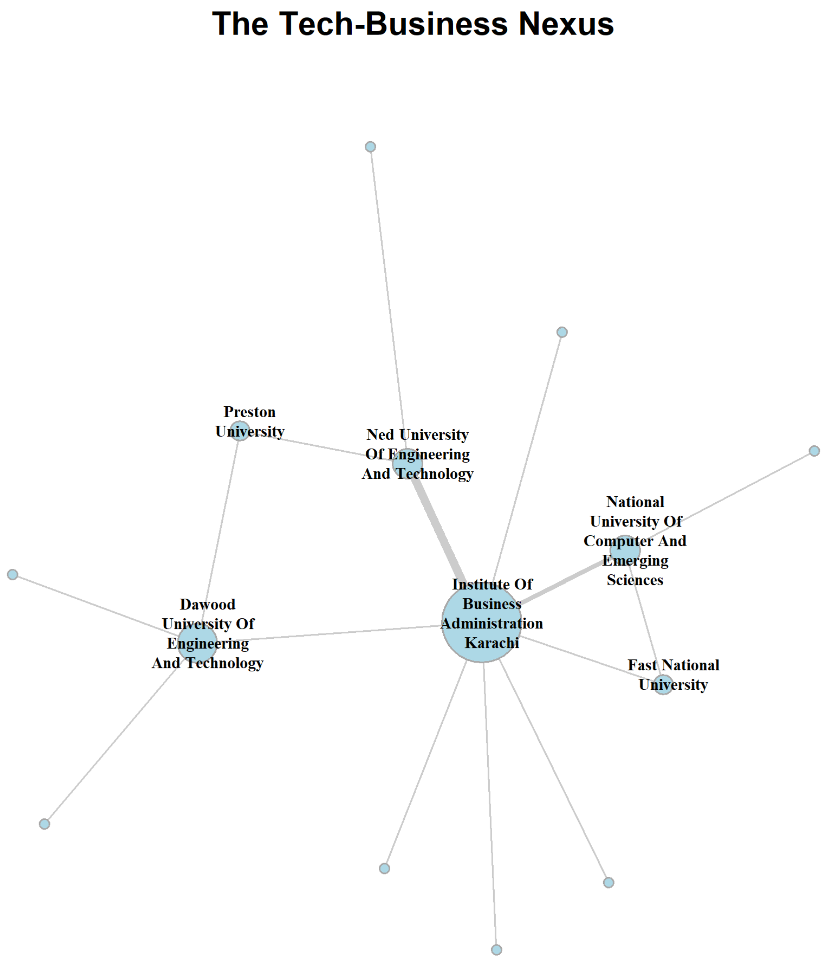

Figure 6: Graph of Cluster A: Tech-Business Nexus.

Cluster B: The Lahore-International Axis

Key Hubs: LUMS, UET Lahore, University of Oxford, University of Cambridge

Unlike the Karachi cluster, the "Lahore pathway" is structurally integrated with international institutions, particularly the British educational elite (Oxbridge) (Figure 7). We theorize this may suggest a pathway rooted in "prestige", where CEOs graduate from Lahore-based institutions and proceed to acquire another degree from an international institution to acquire "legitimacy", which may assist them in climbing the corporate ladder back in Pakistan.

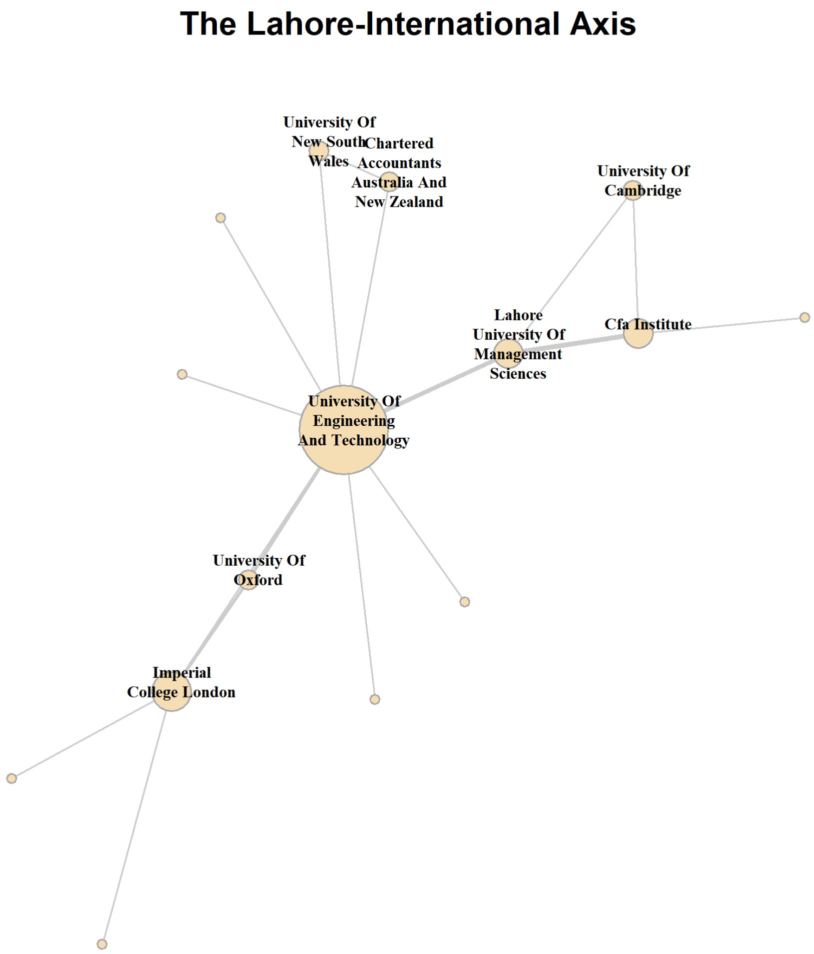

Figure 7: Graph of Cluster B: Lahore-International Axis.

Cluster C: The National Legacy and Accounting Cluster

Key Hubs: Punjab University, Karachi University, ICAP (Chartered Accountants)

Cluster C acts as the 'National Bridge,' uniting the country's two oldest public-sector giants—Punjab University and Karachi University (Figure 8), located in the two educational hubs of Pakistan, Lahore and Karachi respectively. This community represents a "Traditional Financial-Professional Pathway". Unlike the modern MBA or tech routes, this cluster is defined by a combination of legacy public education and professional accreditation by institutions such as ICAP. This highlights a distinct technocratic road to the CEO office, where leadership legitimacy is derived from demonstrable technical competence (Professional charter) rather than private-school prestige.

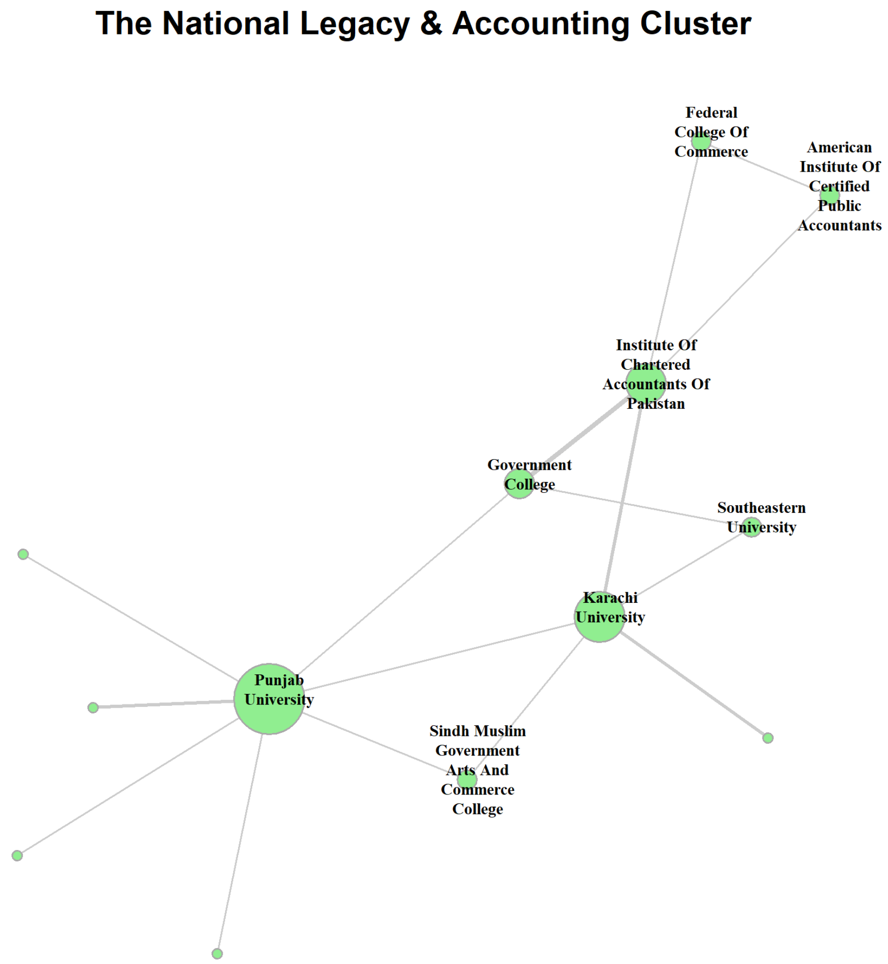

Figure 8: Graph of Cluster C: National Legacy and Accounting Cluster.

Cluster D: The UK Cluster

Key Hubs: London School of Economics (LSE), Institute of Chartered Accountants in England & Wales (ICAEW), University of London, London Metropolitan University

While Cluster C represents the indigenous financial-technocratic route, Cluster D (Figure 9) embodies the "Cosmopolitan Financial" pathway. Concentrated in London and anchored by the London School of Economics (LSE) and the ICAEW, this cluster mirrors the professional-accreditation focus of the National Legacy group but operates on a global scale. Crucially, it is also distinct from the Lahore-International Axis (Cluster B). Unlike the "Collegiate Prestige" of the Oxbridge system, which relies on historic traditions and social capital, this cluster focuses on "Specialized Expertise" where leadership legitimacy is derived from mastery in navigating the financial ecosystem of the City of London.

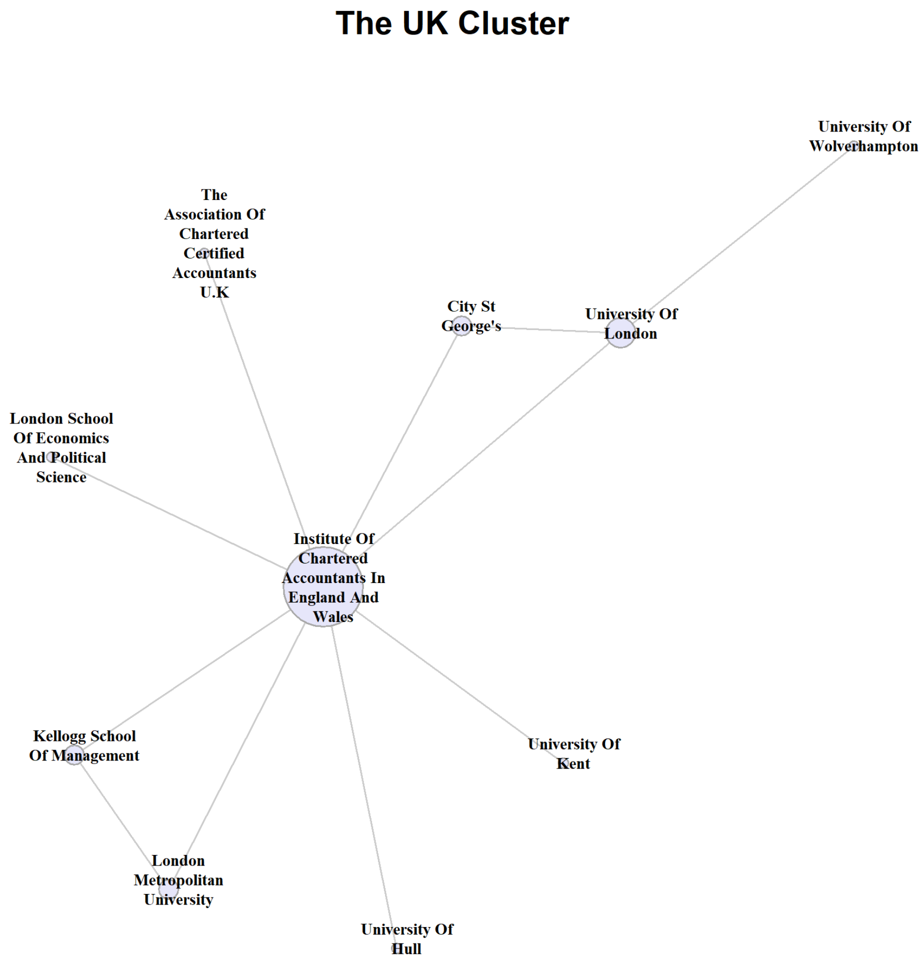

Figure 9: Graph of Cluster D: UK Cluster.

5.2.2 Node-Level Analysis

While community detection reveals the network's divisions, centrality analysis identifies the distinct structural roles played by specific institutions within those divisions. Our analysis focuses on three key metrics: Degree (Volume), Betweenness (Brokerage), and PageRank/Eigenvector (Systemic Influence).

5.2.3 Degree Centrality

In the original bipartite network, degree centrality measured volume (how many CEOs an institution produced), identifying "Feeders" like IBA. However, in this unipartite projection, degree centrality takes on a different meaning. Here, a connection exists between two universities only if a CEO has attended both. Therefore, a high degree in this graph indicates that an institution acts as a "Mixer", i.e. it produces CEOs which go on to attend different Institutions. IBA leads with a degree of 13, meaning

Rank	Degree	Eigenvector	PageRank	Betweenness
1	IBA	IBA	IBA	PU
2	UET	NED	UET	IBA
3	PU	KU	ICAEW	UET
4	ICAEW	NUCES	PU	London Met
5	KU	LUMS	KU	GCU
6	NED	UET	NED	Cambridge
7	ICAP	PU	ICAP	ICAEW
8	LUMS	CFA	LUMS	SOAS
9	GCU	ICAP	INSEAD	UPenn
10	CFA	SEU	QAU	ICAP

Note: **IBA**: Institute of Business Administration; **UET**: University of Engineering & Technology (Lahore); **PU**: Punjab University; **KU**: Karachi University; **NED**: NED University of Engineering & Technology; **LUMS**: Lahore University of Management Sciences; **ICAP**: Institute of Chartered Accountants of Pakistan; **ICAEW**: Institute of Chartered Accountants in England & Wales; **NUCES**: National University of Computer & Emerging Sciences (FAST); **GCU**: Government College University, Lahore; **CFA**: CFA Institute; **SEU**: Southeastern University; **QAU**: Quaid-e-Azam University; **London Met**: London Metropolitan University; **Cambridge**: University of Cambridge; **SOAS**: SOAS University of London; **UPenn**: University of Pennsylvania.

Table 1: Centrality Measures for Top Institutions

it is directly connected to 13 other unique institutions via shared alumni. UET Lahore follows closely with 12.

Significantly, if IBA were only a “feeder”, it would produce many CEOs but remain isolated if those CEOs never went anywhere else. Its high "mixer" score proves that IBA alumni are highly mobile, frequently holding second degrees from diverse institutions such as engineering schools or foreign universities. This makes IBA a central hub for cross-pollinating corporate culture.

5.2.4 Betweenness Centrality

Betweenness centrality identifies “bridges” that connect separate clusters. In this metric, Punjab University surprisingly overtakes IBA for the top spot. Thus, while IBA produces the most CEOs, Punjab University serves as the network’s critical structural bridge. As the anchor of the “National Legacy” cluster, it connects the “Legacy” of the public sector with the modern “Tech-Business” elites of Karachi and Lahore.

International institutions like London Metropolitan University and University of Cambridge appear more frequently in the top 10 for Betweenness despite having lower degree centrality. This proves their role as “Weak Ties”, i.e. they may serve as international neutral grounds where CEOs from different Pakistani clusters converge, facilitating information flow between otherwise segregated groups.

5.2.5 PageRank vs. Eigenvector Centralities

Eigenvector centrality is heavily skewed towards the Karachi-based "Tech-Business Nexus", with the top 3 spots occupied by IBA, NED, and Karachi University. This suggests a dense, self-reinforcing network structure within the Karachi educational community. The high frequency of shared alumni between these institutions, combined with the dominance of IBA, creates a feedback loop that mathematically amplifies their collective Eigenvector scores.

PageRank, which measures holistic influence across the entire network, corrects this bias. In the PageRank listings, UET Lahore rises to Rank 2, and influence is generally more evenly distributed between Karachi and Lahore-based institutions, perhaps evidenced by PageRank more closely mirroring the Degree centrality ranking.

This suggests that while Karachi-based institutions are more structurally cohesive (Higher Eigenvector), the systemic influence of the corporate network is balanced between the two major cities (Balanced PageRank).

5.3 CEO-to-CEO Projection

While the primary focus of this paper is to identify institutional pathways to power, a brief analysis of the projected CEO-CEO network is necessary to understand the social cohesion of the corporate elite (Figure 10). In this projection, two CEOs are connected if they share an alma mater. We will limit this analysis to global network metrics and node-level centrality to establish the connectivity of the social graph, and we will mention communities without going into in-depth community analysis.

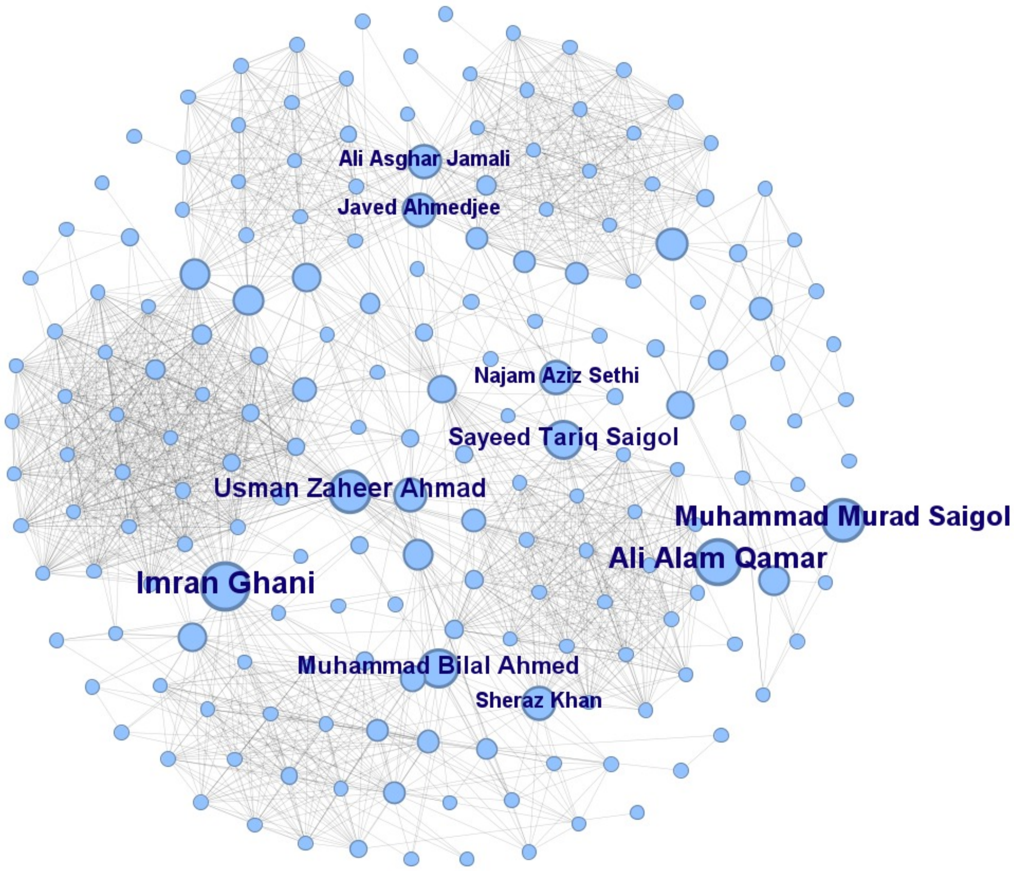

Figure 10: Giant Component of the CEO-CEO Projection.

Though the graph is sparse with a density of **0.068**, the giant component of the network consists of **80%** of the total nodes. This giant component exhibits Small World properties, characterized by an extremely high Average Clustering Coefficient **(0.88)** and a relatively short Average Path Length **(3.17)**. This indicates that the KSE-500 leadership may form a dense core **"Old Boys Club"**, where information, reputation and social capital travels rapidly across the network and most CEOs only have three degrees of separation.

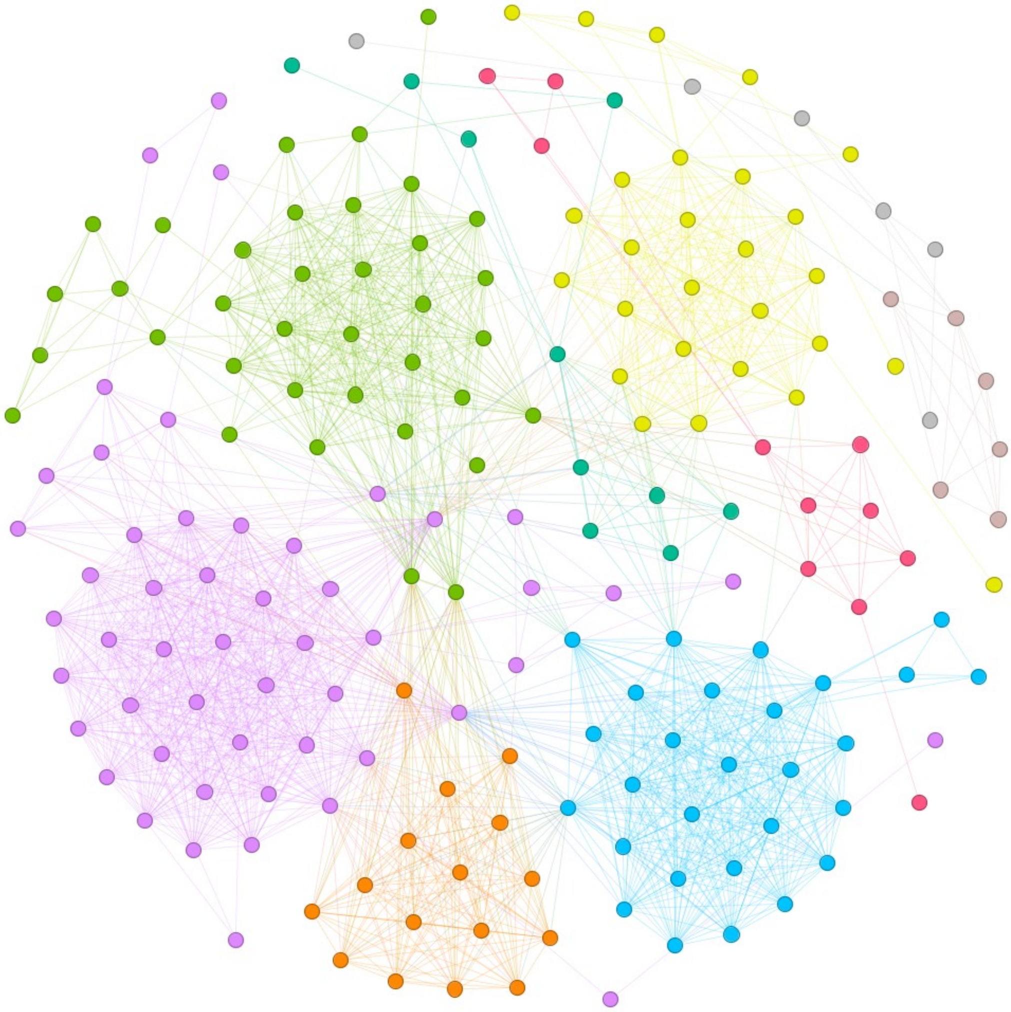

Figure 11: Communities within the CEO-CEO Projection.

By applying the Label Propagation algorithm, we discovered that the giant component consists of **9** communities, confirming its fragmented structure (Figure 11). The detailed analysis of these communities, however, goes beyond the scope of this paper.

5.3.1 Node-Level Analysis

Rank	Degree	Eigenvector	PageRank	Betweenness
1	Usman Zaheer Ahmad	Usman Zaheer Ahmad	Imran Ghani	Imran Ghani
2	Imran Ghani	Syed Masood Abbas Jaffery	Usman Zaheer Ahmad	Ali Alam Qamar
3	Syed Masood Abbas Jaffery	Amir R. Paracha	Amir R. Paracha	Muhammad Murad Saigol
4	Amir R. Paracha	Imran Ghani	Syed Masood Abbas Jaffery	Usman Zaheer Ahmad
5	Muhammad Akram Shahid	Yousaf Mirza	Muhammad Akram Shahid	Muhammad Bilal Ahmed
6	Javed Ahmedjee	Azfar Arshad	Ali Asghar Jamali	Sayeed Tariq Saigol
7	Ali Asghar Jamali	Kamran Arshad Inam	Javed Ahmedjee	Najam Aziz Sethi
8	Yousaf Mirza	Syed Muhammad Daniyal	Syed Jarar Haider Kazmi	Javed Ahmedjee
9	Kamran Arshad Inam	Jamal Nasir Khan	Fahd Kamal Chinoy	Ali Asghar Jamali
10	Azfar Arshad	Mujtaba Ahmad	Faheem Haider	Sheraz Khan

Table 2: Centrality Measures for Top CEOs

5.3.2 Degree Centrality

Degree centrality measures the number of peers a CEO shares an alma mater with. Usman Zaheer Ahmad (CEO, Fauji Foods) ranks the highest, followed closely by Imran Ghani (CEO, KSB Pumps). These high-degree actors are deeply embedded in the giant component, and their connectivity suggests they have the fastest access to information in elite circles.

5.3.3 Betweenness Centrality

Betweenness identifies "Gatekeepers"—CEOs who sit on the shortest path between other pairs of CEOs. These actors are often critical for moving novel information between otherwise disconnected groups.

Here, Imran Ghani overtakes Usman Zaheer Ahmad for the top spot. While Usman Zaheer has more "friends" (Degree), Imran Ghani connects "different groups of friends". This "Brokerage" position gives structural bridges significant strategic power to control the flow of information and social capital in the network.

5.3.4 PageRank vs. Eigenvector Centralities

These are measures of "influence" in the network. Eigenvector centrality, which favors actors located in dense cliques, has Usman Zaheer at the top. However, the metric is saturated, with the top 10 CEOs all scoring near a perfect 1. This supports the idea that the core cluster is an "echo chamber", and the underlying structure is self-reinforcing. PageRank, which favors holistic influence, breaks the tie, placing Imran Ghani at the top. So, while Usman Zaheer leads in local (Core) influence, Imran Ghani leads in global (Systemic) influence.

5.4 Comparison with Seminal Network Models (Institution Projection)

To validate the structural properties of the Institution-Institution network, we compared the metrics of its Giant Component against three seminal theoretical models: Erdos-Renyi, Barabasi-Albert, and Watts-Strogatz.

5.4.1 Graph Model Metrics

Graph Model	Average Path Length	Clustering Coefficient	Follows Power Law?
Our Graph (Giant Component)	4.18	0.107	Yes
Erdos-Renyi (Random)	4.10	0.045	No
Barabasi-Albert (Scale-Free)	5.22	0.000	Yes
Watts-Strogatz (Small World)	3.15	0.429	No

Table 3: Comparison of Empirical Network vs. Theoretical Models

5.4.2 Scale-Free Property

The analysis confirms that the real network follows a Power Law distribution which is consistent with the Barabasi-Albert model. In contrast, the random Erdos-Renyi graph does not. This mathematically validates the "Rich-Get-Richer" dynamic. The network is not random, but is dominated by a few "super-hubs". This suggests that CEO aspirants do not randomly decide upon their educational institution, rather they deliberately choose institutes that have a greater amount of CEO alumni.

5.4.3 Small World Property

The Average Path Length (4.18) of the Giant Component is remarkably similar to that of a random graph (4.09). However, the Clustering Coefficient (0.107) is more than double that of the random graph (0.045). This combination suggests that the institutions are locally dense yet still globally connected. This validates our finding that the network exists as fragmented islands, while also confirming that the Institute network operates as a “Small World”, i.e. at maximum, there are only 4 degrees of separation between any two institutes.

6 Results Summary

Bipartite Structure: The network is defined by extreme sparsity (0.004) and follows a Power Law, confirming it is Scale-Free. This indicates a "Rich-get-Richer" dynamic where a few "super-hubs" (IBA, Punjab University) dominate the production of CEOs.

This structural inequality suggests that access to the corporate elite is highly centralized, which can pose a barrier to entry for candidates outside these specific institutional pipelines.

Institution-Institution Projection: The network is characterized by a low global density (0.176) but a significantly higher local clustering coefficient (0.38), mathematically confirming a structure fragmented into distinct “islands”, which exhibit "small world" properties. Community detection identifies four specific power blocs defined by geography and methodology:

- **Tech-Business Nexus (Karachi):** A dense, self-reinforcing cluster of business schools such as IBA and engineering schools such as NED.
- **Lahore-International Axis:** A prestige-driven pathway linking Lahore-based institutes with international elite universities such as Oxbridge.
- **National Legacy Cluster:** An indigenous technocratic route anchored by public sector giants and professional certification bodies (ICAP).
- **UK Cluster:** A cluster of finance-oriented institutions centered around the financial ecosystem of the City of London.

While IBA leads in sheer volume (**Degree Centrality**) and acts as the primary "mixer" for cross-pollinating corporate talent, Punjab University and international schools serve as critical “bridges” (**Betweenness Centrality**). International institutions further function as "Weak Ties", neutral grounds where segregated regional clusters can convey novel information. Without these bridges, the network would likely fracture into completely disconnected silos, resulting in significant information bottlenecks in the KSE-500 corporate spheres.

CEO-CEO Projection: The giant component of the CEO network exhibits strong "small world" properties and an exceptionally high Average Clustering Coefficient of 0.88, significantly higher than a random graph. This high level of transitivity, where a friend of a friend is almost always a friend, indicates that the corporate elites form a highly cohesive social core. The structural properties highlight the power of “bridges” to act as controllers of information and social capital, while the saturation of high Eigenvector centrality scores further suggests a self-reinforcing structure. Thus, while the underlying institutions are fragmented, the resulting social layer of CEOs operates as a series of overlapping, exclusive cliques. This mirrors the "Small Club" characterization found in existing literature, suggesting that shared educational backgrounds serve as a powerful mechanism for social closure and elite reproduction in Pakistan.

7 Discussion of Limitations and Future Direction

A significant challenge for this project was the absence of a pre-existing dataset on the educational backgrounds of KSE-500 CEOs. We constructed a custom dataset by compiling information from external sources. However, this data was often incomplete, with some CEOs lacking education details entirely or listing a graduate institution without an undergraduate one. Consequently, we had to exclude records with incomplete information which may introduce a selection bias favoring executives with more prominent digital footprints. Furthermore, the current dataset exists as a static snapshot, providing information only about the current cohort of CEOs in the KSE-500 rather than a temporally complete picture. This may obscure generational shifts in elite recruitment.

To build on this foundation, our future work will focus on improving data completeness. We also plan to differentiate institutions by degree level (Bachelor's, Master's, PhD) and perform community analysis of the CEO network projection in greater depth.

8 Conclusion

The KSE-500 educational network is efficient but insular. It operates as a "Small World" defined by local specialization, where domestic "super-hubs" often act as primary pathways for reaching the CEO office.

The stark division between clusters, such as the "Karachi Nexus" and "Lahore Axis", highlights a geographic and methodological split in leadership development. Though insular, the network avoids total stagnation through specific "bridge" institutions—often international or legacy public universities—that facilitate the flow of information between these clusters. Hence, while domestic institutions serve as the primary gateways to the executive suite, foreign credentials function as essential bridges that unify Pakistan's fragmented corporate elite.

The cohesiveness of the social network created by these institutions suggests that the "Small Club" properties observed in global corporate structures also define the Pakistani sphere. This suggests that alongside biological kinship, sociologically constructed alumni "families" may also propagate elite exclusivity. Ultimately, this structural insularity may incentivize aspirants to deliberately opt for the alma maters of existing CEOs, creating a feedback loop that further entrenches the dominance of these select institutions.

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